

MONTREAL MCTAVISH, QC, Canada

WMO: 716120

Lat: 45.5050N

Lon: 73.5792W

Elev: 73

StdP: 100.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.6	(c) -19.1	(d) -28.3	(e) 0.3	(f) -20.8	(g) -25.6	(h) 0.4	(i) -18.3	(j) 5.5	(k) -9.4	(l) 5.1	(m) -9.2	(n) 2.2	(o) 270	(p) 0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.0	(c) 30.2	(d) 21.8	(e) 28.7	(f) 20.9	(g) 27.3	(h) 20.1	(i) 23.3	(j) 28.2	(k) 22.3	(l) 26.7	(m) 21.4	(n) 25.6	(o) 2.1	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.6	16.4	26.2	20.7	15.5	25.2	19.8	14.7	24.2	69.7	28.2	65.9	26.7	62.5	25.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.9	4.3	3.8	-24.4	32.8	2.8	1.4	-26.4	33.8	-28.0	34.6	-29.6	35.3	-31.6	36.3
			-24.9	25.2	2.6	1.1	-26.8	26.0	-28.3	26.7	-29.8	27.3	-31.7	28.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-7.8	-6.3	-0.9	6.8	14.3	19.4	22.0	21.2	17.0	9.9	3.0	-3.9	(6)
(7)		DBStd	11.55	6.68	5.92	5.71	4.77	4.39	3.69	2.79	2.89	3.99	4.40	4.86	5.83	(7)
(8)		HDD10.0	2182	553	458	339	119	10	0	0	1	58	214	429	(8)	
(9)		HDD18.3	4111	812	691	596	347	139	31	3	8	72	264	461	688	(9)
(10)		CDD10.0	1436	0	0	2	22	144	281	373	346	211	54	4	0	(10)
(11)		CDD18.3	324	0	0	0	1	15	62	118	96	32	2	0	0	(11)
(12)		CDH23.3	1941	0	0	1	8	106	396	737	521	168	4	0	0	(12)
(13)		CDH26.7	436	0	0	0	1	21	96	176	108	35	0	0	0	(13)
(14)	Wind	WSAvg	1.9	2.0	2.0	2.1	2.2	2.0	1.8	1.7	1.6	1.7	1.8	1.8	1.9	(14)
(15)	Precipitation	PrecAvg	1095	73	74	81	80	82	93	101	105	95	85	101	108	(15)
(16)		PrecMax	1379	142	181	137	164	164	151	193	205	190	145	234	203	(16)
(17)		PrecMin	888	19	23	20	24	16	21	41	31	22	20	40	48	(17)
(18)		PrecStd	134	31	40	33	29	38	36	39	45	41	32	37	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	8.2	17.1	24.2	29.1	31.6	32.3	31.4	30.3	23.7	16.8	10.9	(19)
(20)			MCWB	8.4	5.7	12.2	15.8	19.2	22.6	23.5	22.6	21.9	17.4	13.7	9.4	(20)
(21)		2%	DB	5.9	5.6	11.4	19.5	26.2	29.2	30.2	29.2	27.3	20.7	14.0	7.6	(21)
(22)			MCWB	4.9	3.5	6.9	11.6	17.1	21.0	22.0	21.2	20.3	15.7	11.5	6.2	(22)
(23)		5%	DB	3.6	3.7	8.4	16.6	23.8	27.4	28.6	27.7	25.1	18.2	11.8	5.4	(23)
(24)			MCWB	2.3	2.0	4.9	10.0	15.6	19.8	21.2	20.5	19.4	14.0	9.3	4.0	(24)
(25)		10%	DB	1.4	2.0	6.4	13.9	21.5	25.6	27.2	26.3	23.2	16.2	9.7	3.5	(25)
(26)			MCWB	0.2	0.6	3.4	8.2	14.7	18.4	20.2	19.8	18.5	12.6	7.2	2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	6.3	13.0	16.7	20.8	23.7	25.0	24.1	23.1	19.0	14.5	9.6	(27)
(28)			MCDWB	9.2	7.9	16.7	22.6	26.6	29.9	30.8	29.3	28.0	22.4	15.8	10.8	(28)
(29)		2%	WB	4.9	3.8	7.6	13.3	19.0	22.3	23.6	22.7	21.5	16.4	11.8	6.3	(29)
(30)			MCDWB	5.8	5.3	10.4	17.5	23.8	27.7	28.3	27.2	25.6	19.7	13.8	7.5	(30)
(31)		5%	WB	2.3	2.2	5.4	10.8	17.4	21.0	22.5	21.7	20.2	14.6	9.6	4.1	(31)
(32)			MCDWB	3.5	3.5	8.0	15.3	21.9	25.4	26.7	25.7	24.2	17.6	11.4	5.2	(32)
(33)		10%	WB	0.4	0.7	3.7	8.9	15.6	19.9	21.4	20.9	19.0	13.1	7.4	2.3	(33)
(34)			MCDWB	1.4	2.0	6.0	13.5	19.9	24.2	25.7	24.7	22.7	15.6	9.4	3.4	(34)
(35)	Mean Daily Temperature Range	MDBR		7.0	7.1	7.1	8.1	8.8	8.4	8.0	8.0	6.7	5.8	5.8	(35)	
(36)		5% DB	MCDBR	8.3	7.9	9.3	12.3	11.4	10.5	9.2	9.2	9.6	9.4	8.7	7.2	(36)
(37)			MCWBR	8.3	7.2	6.8	7.6	6.1	5.6	4.8	4.5	5.2	6.0	7.6	6.9	(37)
(38)		5% WB	MCDWB	8.3	7.7	8.9	11.2	9.9	9.0	8.2	8.3	8.8	8.6	8.5	7.1	(38)
(39)			MCWBR	8.5	7.4	7.3	7.9	6.5	5.6	5.1	4.8	5.4	6.3	7.9	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.258	0.288	0.330	0.362	0.388	0.439	0.428	0.413	0.373	0.335	0.306	0.292	(40)
(41)		taud		2.348	2.257	2.292	2.360	2.345	2.207	2.245	2.282	2.383	2.494	2.489	2.306	(41)
(42)		Ebn at Noon		861	891	897	897	880	833	837	836	843	834	799	763	(42)
(43)		Edh at Noon		74	100	112	115	121	140	133	124	103	80	67	71	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.44	2.40	3.64	4.37	5.12	5.63	5.66	5.01	3.96	2.29	1.39	1.00	(44)
(45)		RadStd		0.16	0.25	0.30	0.51	0.48	0.43	0.42	0.23	0.33	0.20	0.18	0.09	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page